
Signals & System

Lecture 1

Introduction to Signals & Variables

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Books / Resources

Essential

- A V Oppenheim, A S Willsky: Signals and Systems, 2nd Ed
- D Hanselman, B Littlefield “Mastering Matlab 6: A comprehensive tutorial and reference”

Recommended

- Haykin “Signals and Systems, John Wiley and Sons, 2002
- <http://www.mit.edu/~6.003/> - Signals and Systems at MIT
- <http://dynamo.ecn.purdue.edu/~bouman/ee301/> - Signals and Systems at Purdue
- <http://www.jhu.edu/~signals/index.html> - on-line set of Java applets demonstrating various signals and system concepts

Course Syllabus

- 1. Concepts** : Systems, signals, mathematical models. Continuous-time and discrete-time signals. Energy and power signals. Linear systems. Examples for use throughout the course, use of Matlab
- 2. Linear systems, Convolution** : Impulse response, input signals as continuum of impulses. Convolution, discrete-time and continuous-time properties
- 3. Basis functions** : Concept of basis function. Fourier series representation of time functions. Fourier transform and its properties. Examples, transform of simple time functions.
- 4. Sampling Discrete-time systems** : Sampling theorem, discrete Fourier transform

Course Syllabus (2)

5. **Laplace transform** : Laplace transform and Fourier transform with convergence factor. Properties of the Laplace transform
6. **Transfer Function of Continuous-Time Systems** : Transfer function, frequency response. Physical realizability, stability. Poles and zeros.
7. **Transfer Function of a Discrete-Time Systems** : Impulse sampler, Laplace transform of impulse sequence, z transform. Properties of the z transform. Examples. Difference equations and differential equations. Digital filters.

Grading Policy

- Quizzes : 10%
- Assignments : 10%
- Sessionals : 30%
- Final Exam : 50%

What is a Signal?

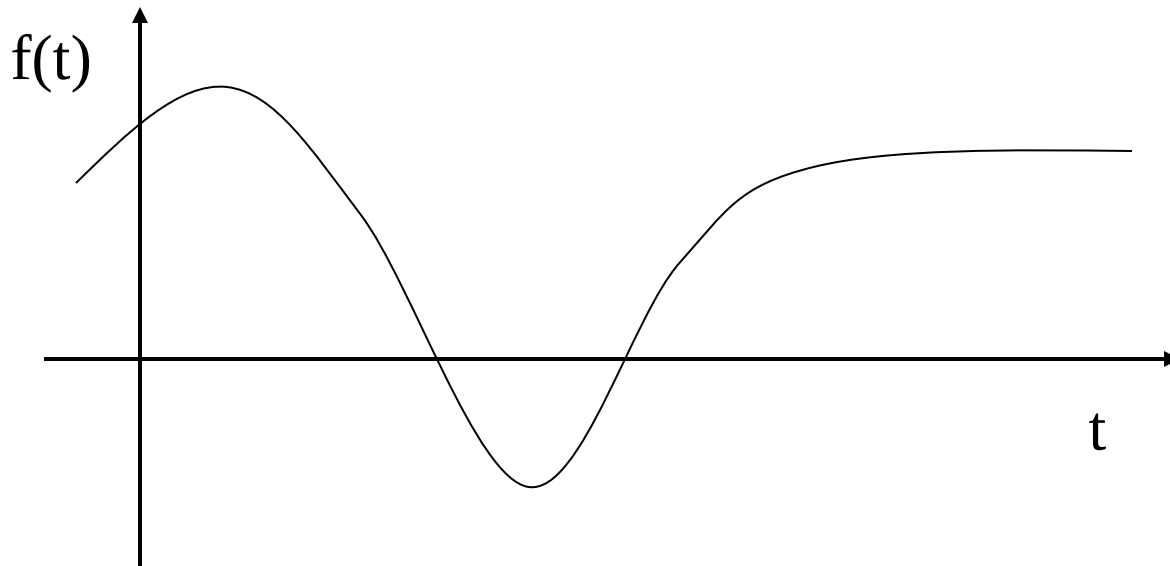
- A signal is a pattern of variation of some form
- Signals are variables that carry information

Examples of signal include:

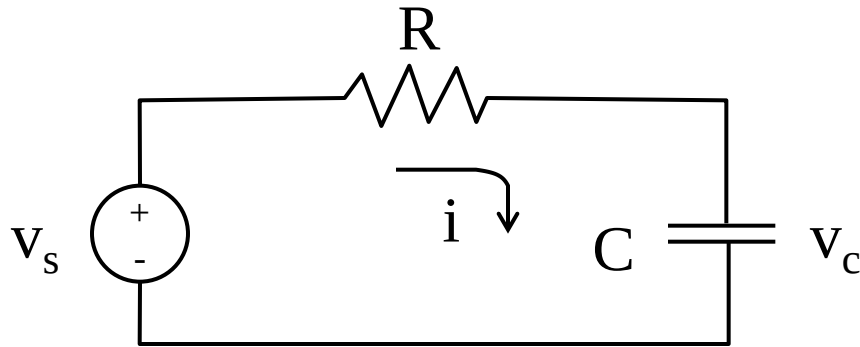
- Electrical signals : Voltages and currents in a circuit
- Acoustic signals: Acoustic pressure (sound) over time
- Mechanical signals: Velocity of a car over time
- Video signals: Intensity level of a pixel (camera, video) over time

How is a Signal Represented?

- Mathematically, signals are represented as a function of one or more **independent variables**.
- For instance a black & white video signal intensity is dependent on x, y coordinates and time t $f(x,y,t)$
- On this course, we shall be exclusively concerned with signals that are a function of a single variable: time



Example: Signals in an Electrical Circuit

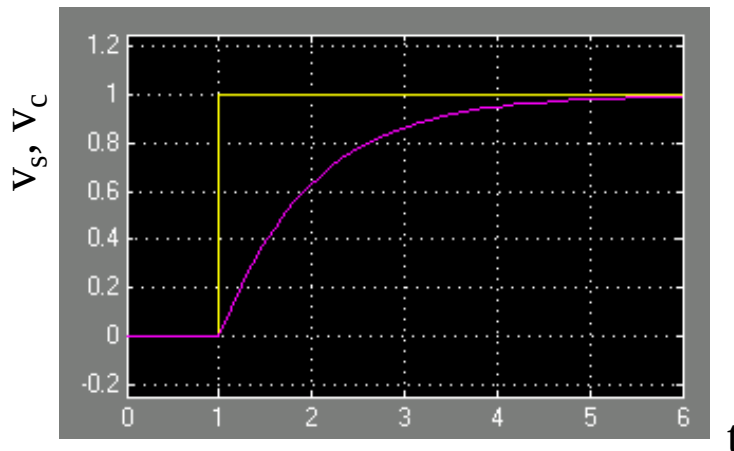


$$i(t) = \frac{v_s(t) - v_c(t)}{R}$$

$$i(t) = C \frac{dv_c(t)}{dt}$$

$$\frac{dv_c(t)}{dt} + \frac{1}{RC} v_c(t) = \frac{1}{RC} v_s(t)$$

- The signals v_c and v_s are patterns of variation over time



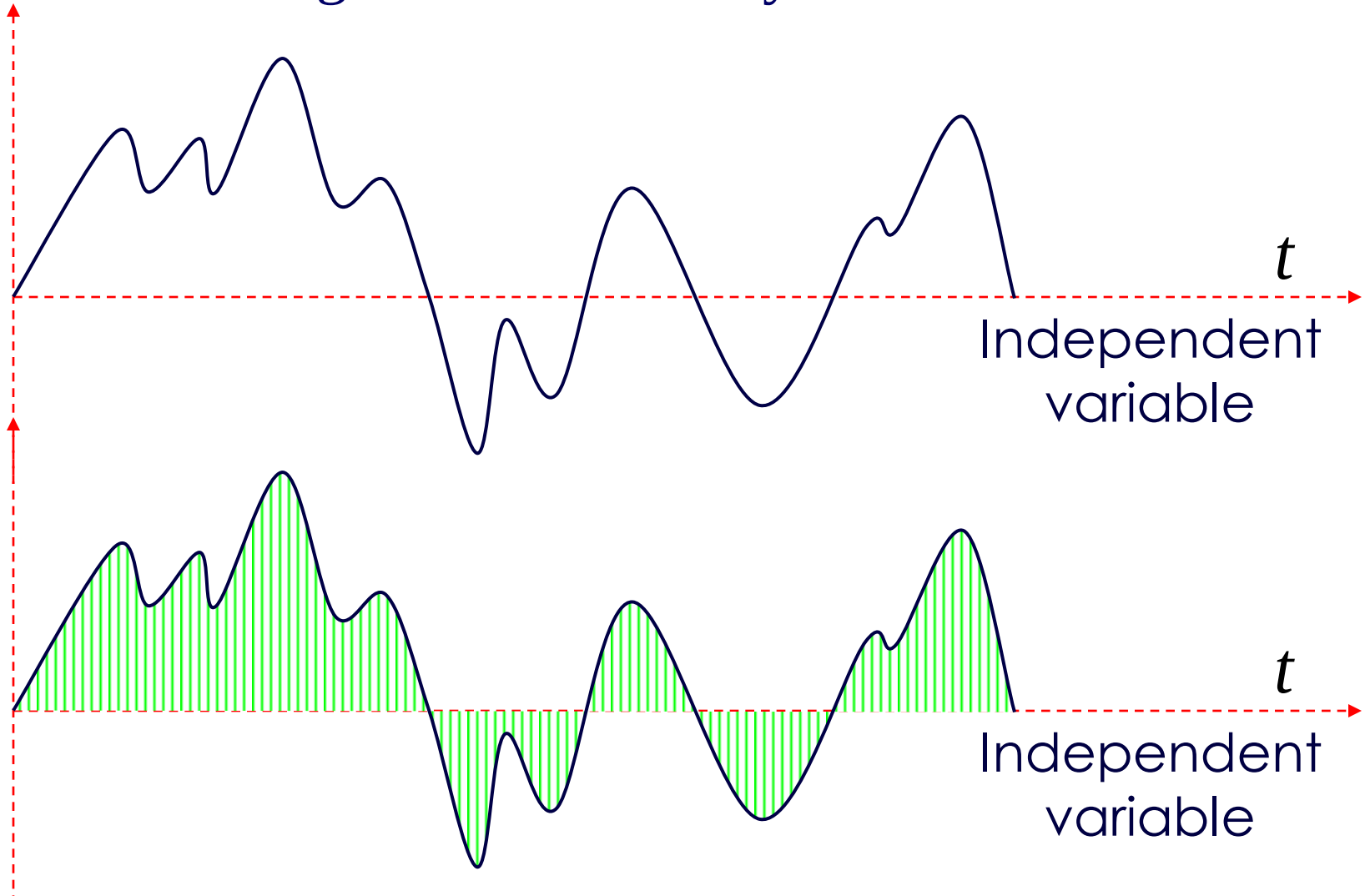
Step (signal) v_s at $t=1$
 $RC = 1$

First order (exponential)
response for v_c

- Note, we could also have considered the voltage across the resistor or the current as signals

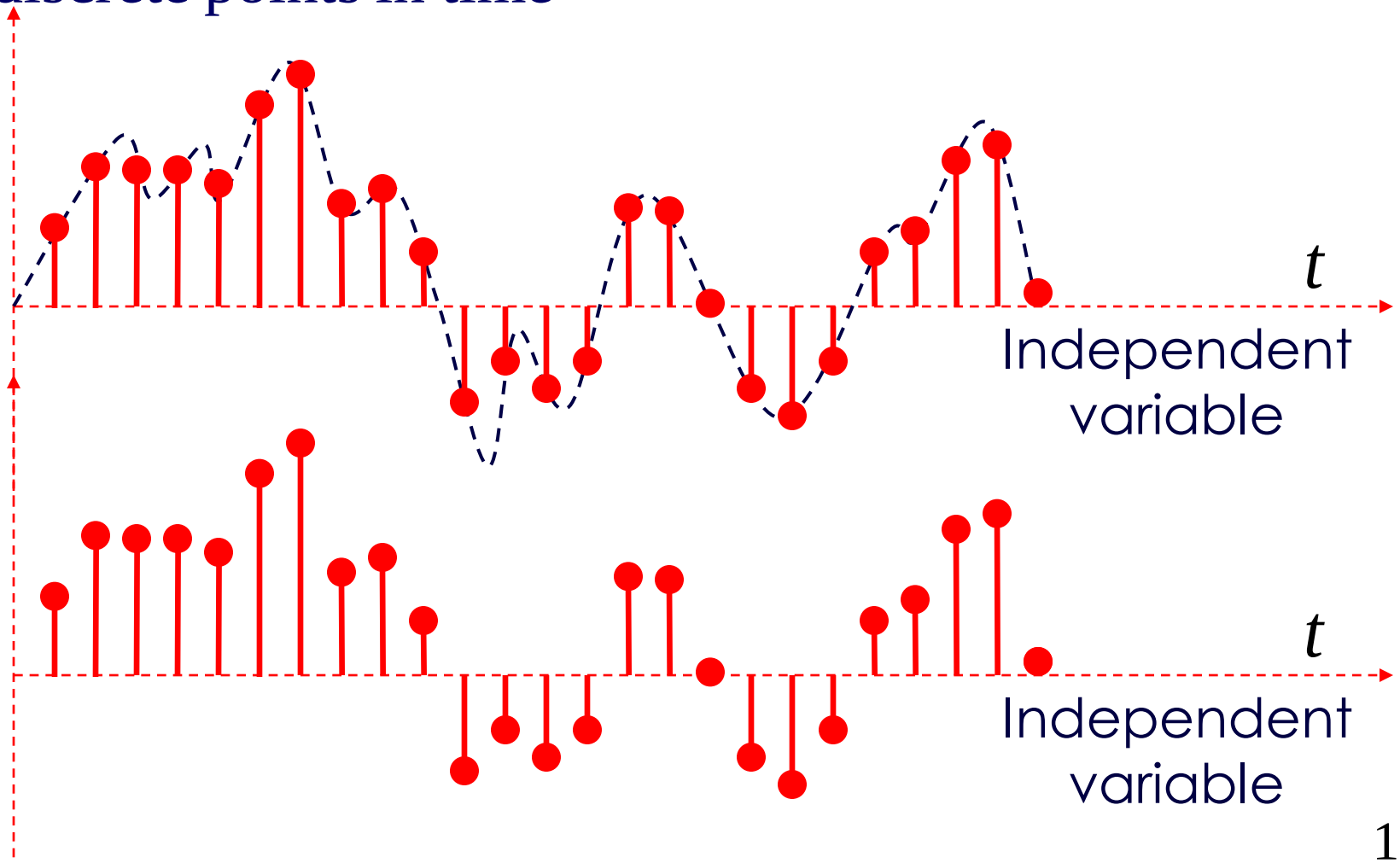
Continuous-time signals

- A value of signal exists at every instant of time



Discrete-time signals

- The value of signal exists only at equally spaced discrete points in time

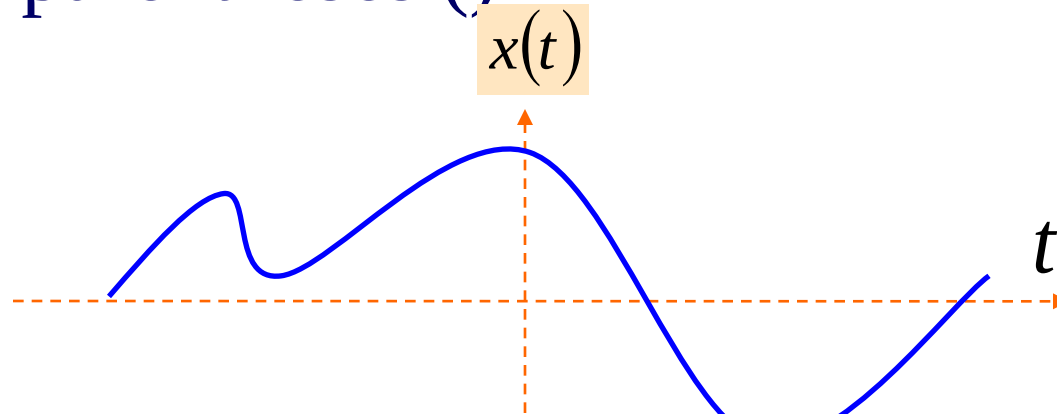


Discrete-time signals

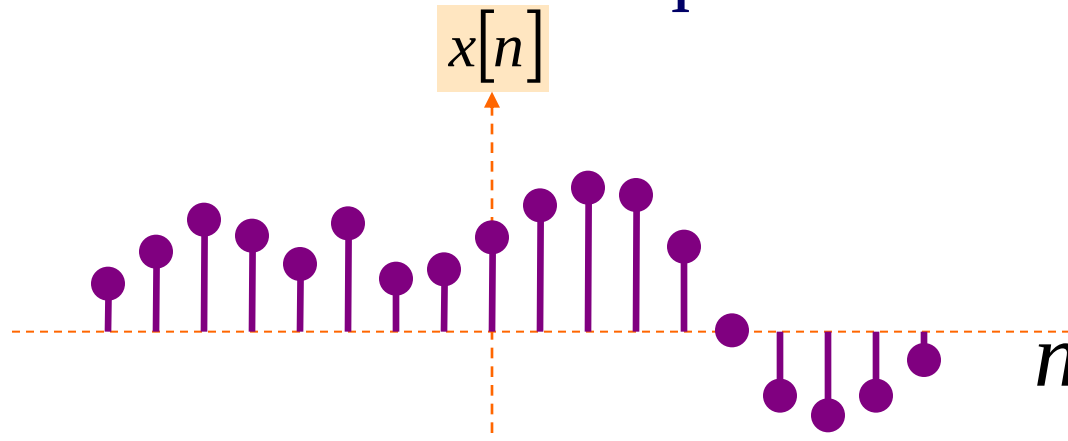
- Why to discretize
- How to discretize
 - How closely spaced are the samples
- Distinction between discrete & digital signals
- How to denote discrete signals
- Is image a discrete or continuous signal
 - The image is generally considered to be a continuous variable
 - Sampling can however be used to obtain a discrete, two dimensional signal (sampled image)

Notation

- A continuous-time signal has independent variable (time) in parentheses ()



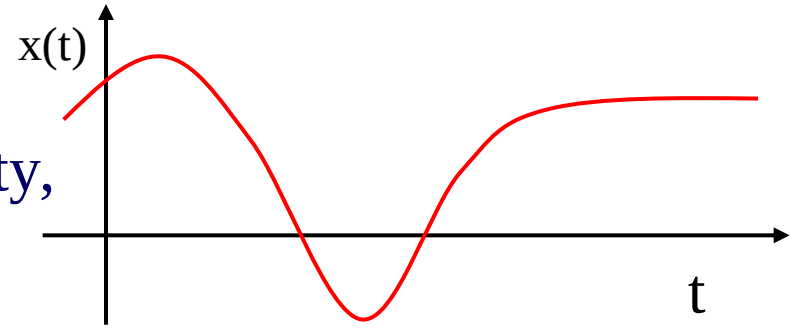
- A discrete-time signal is represented by enclosing the independent variable in square brackets []



Continuous & Discrete-Time Signals

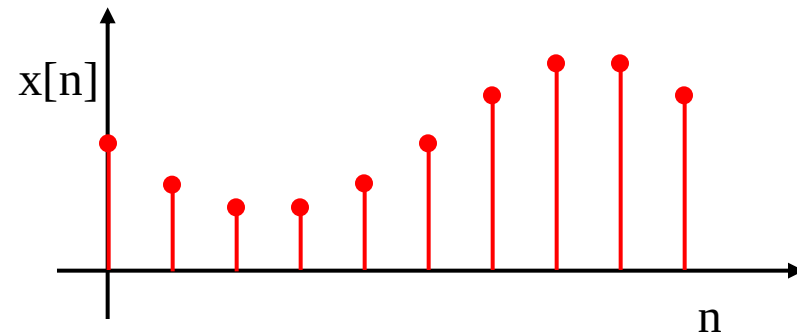
Continuous-Time Signals

- Most signals in the real world are continuous time, as the scale is infinitesimally fine e.g voltage, velocity,
- Denote by $x(t)$, where the time interval may be bounded (finite) or infinite



Discrete-Time Signals

- Some real world and many digital signals are discrete time, as they are sampled e.g. pixels, daily stock price (anything that a digital computer processes)
- Denote by $x[n]$, where n is an integer value that varies discretely



Sampled continuous signal

- $x[n] = x(nk)$ – k is sample time

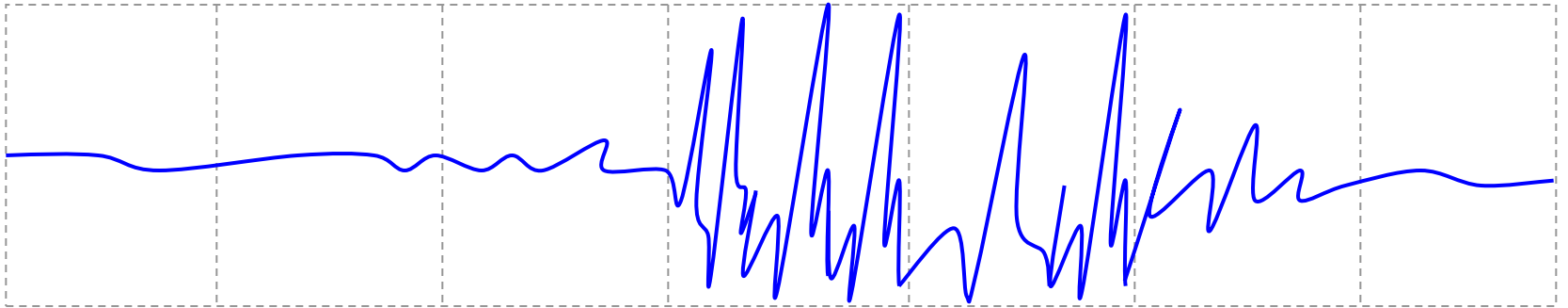
Signal Properties

Particular interest in signals with certain properties:

- **Periodic signals:** a signal that repeats itself after a fixed period T , i.e. $x(t) = x(t+T)$ for all t . e.g. $A \sin(t)$.
- **Even and odd signals:** even if $x(-t) = x(t)$, and odd if $x(-t) = -x(t)$. Examples are $\cos(t)$ and $\sin(t)$ signals.
- **Exponential and sinusoidal signals:** a signal is (real) exponential if it can be represented as $x(t) = Ce^{at}$. The same example is (complex) exponential C and a are complex.
- **Step and pulse signals:** A pulse signal is one which is nearly completely zero, apart from a short spike, $d(t)$. A step signal is zero up to a certain time, and then a constant value after that time, $u(t)$.

Signal

- The Speech Signal



- The ECG Signal



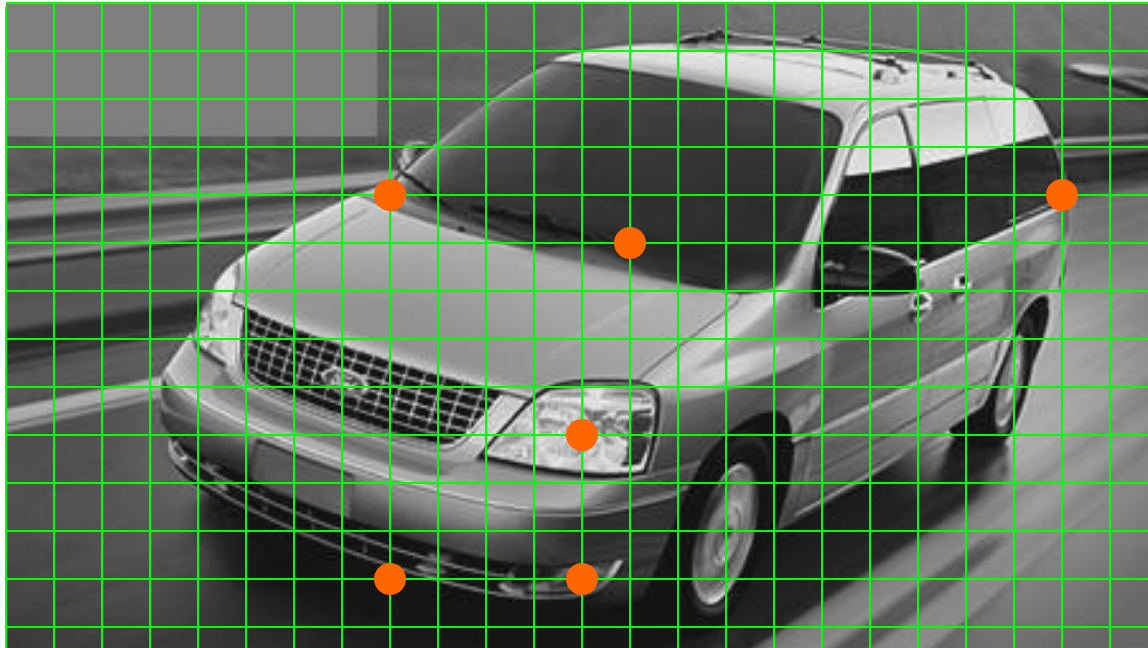
Signal

- The image



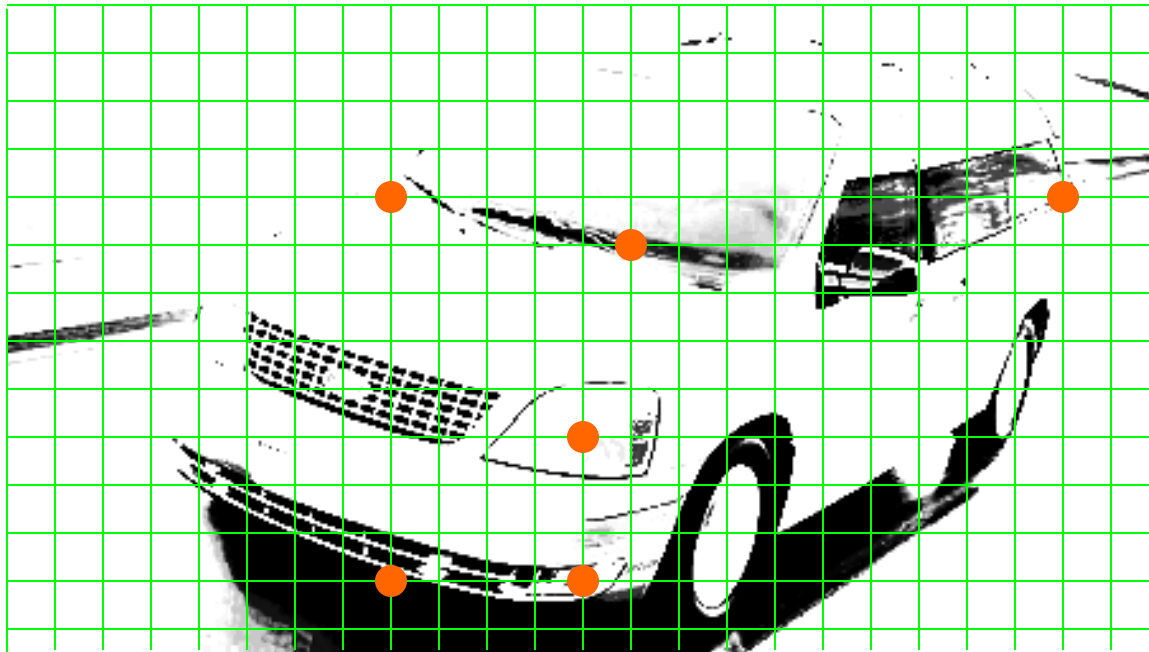
Signal

- The image



Signal

- It is the variation pattern that conveys the information, in a signal



- Signal may exist in many **forms** like acoustic, image, video, electrical, heat & light signal

What is a System?

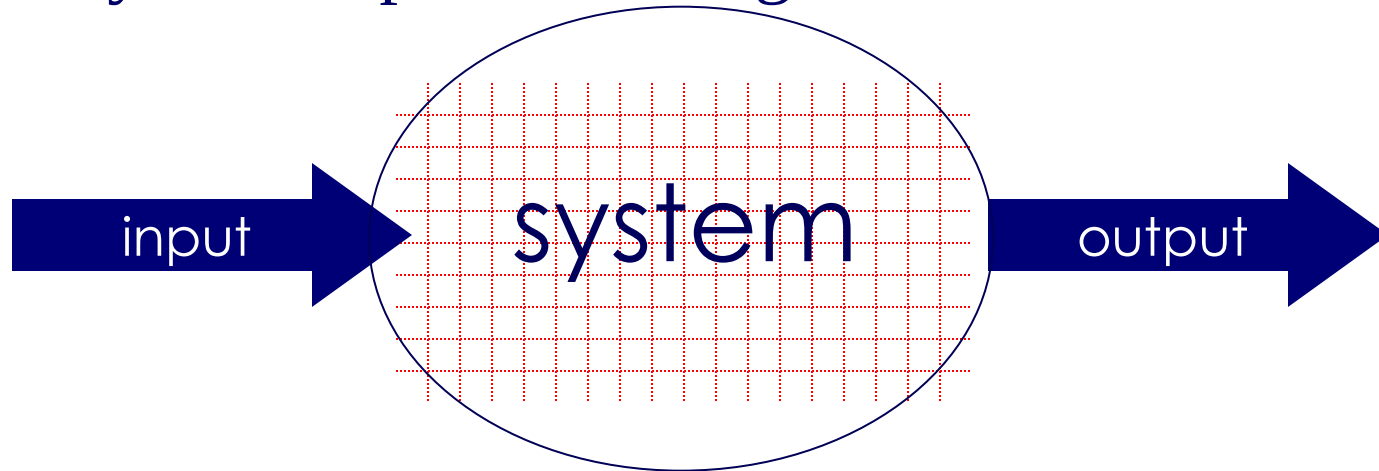
- Systems process input signals to produce output signals

Examples:

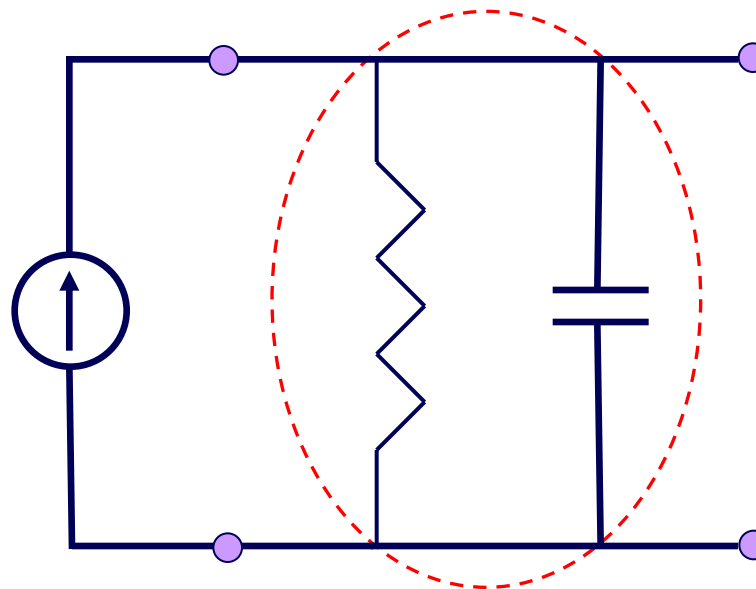
- A circuit involving a capacitor can be viewed as a system that transforms the source voltage (signal) to the voltage (signal) across the capacitor
- A CD player takes the signal on the CD and transforms it into a signal sent to the loud speaker
- A communication system is generally composed of three sub-systems, the transmitter, the channel and the receiver. The channel typically attenuates and adds noise to the transmitted signal which must be processed by the receiver

System

- An entity that responds to a signal



- Examples
 - Circuit



System

- The camera



- The Speech Recognition System



System

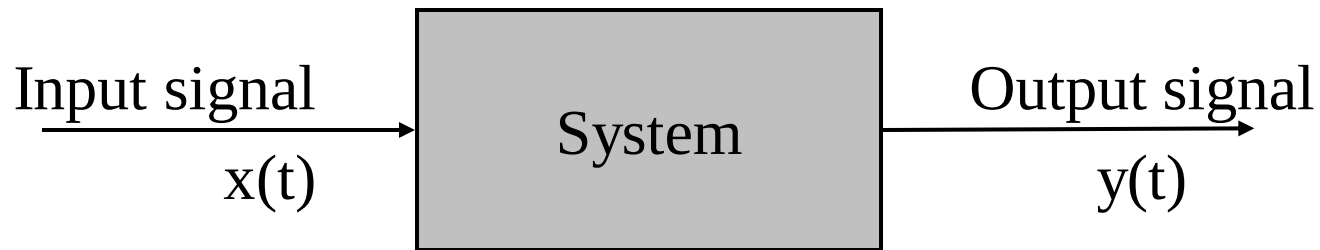
- The audio CD-player
- Block Diagram representation of a system
 - Visual representation of a system



- Shows inter-relations of many signals involved in the implementation of a complex system
- Look at everything around and try to identify the signals and systems !!

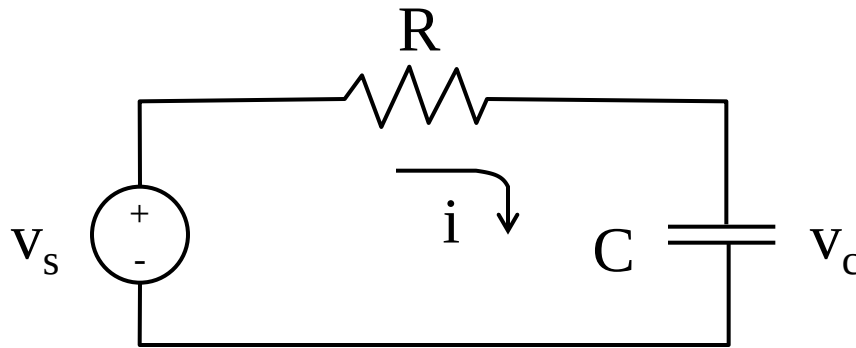
How is a System Represented?

- A system takes a signal as an input and transforms it into another signal



- In a very broad sense, a system can be represented as the ratio of the output signal over the input signal
 - That way, when we “multiply” the system by the input signal, we get the output signal
 - This concept will be firmed up in the coming weeks

Example: An Electrical Circuit System

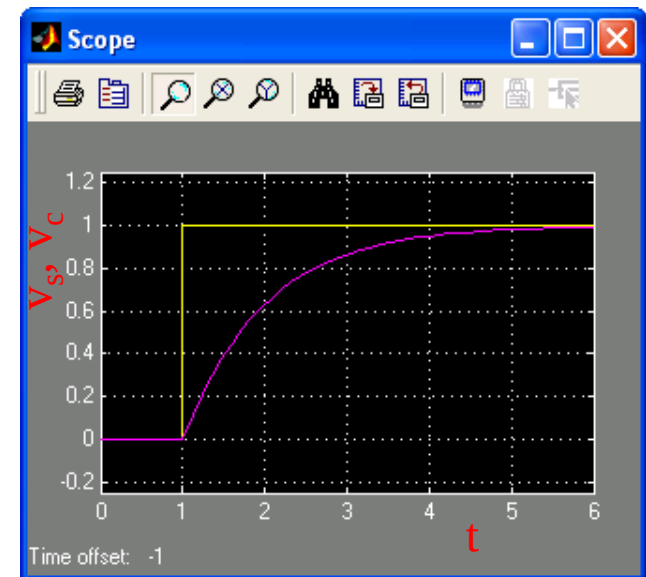
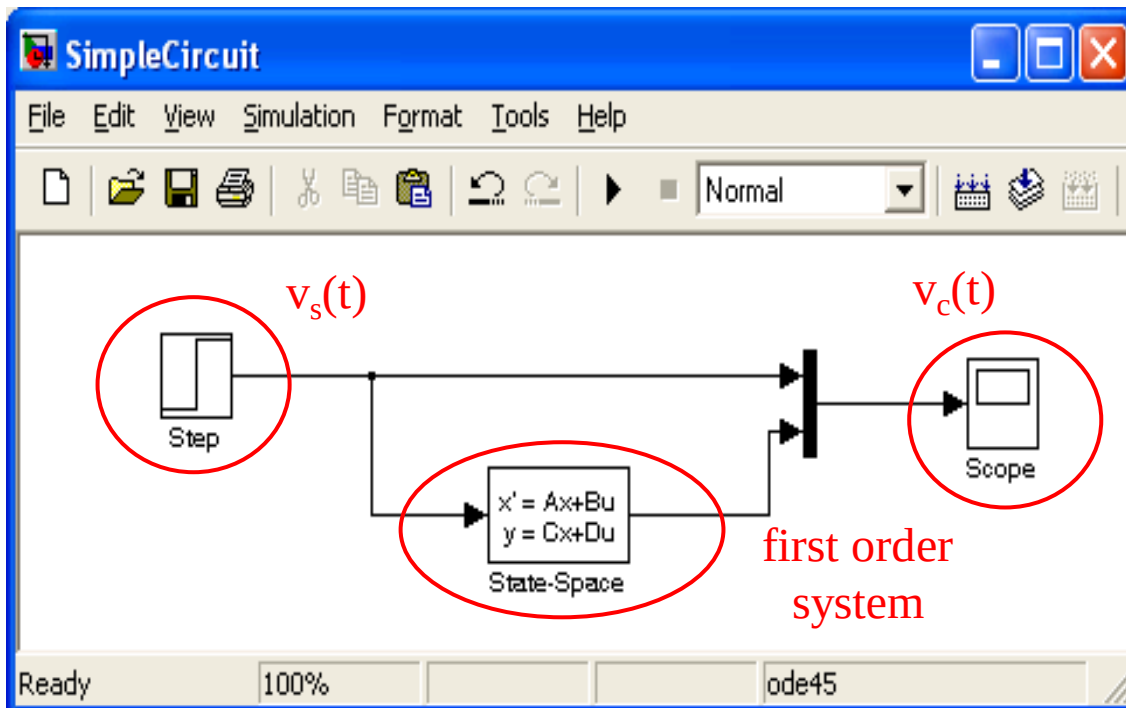


$$i(t) = \frac{v_s(t) - v_c(t)}{R}$$

$$i(t) = C \frac{dv_c(t)}{dt}$$

$$\frac{dv_c(t)}{dt} + \frac{1}{RC} v_c(t) = \frac{1}{RC} v_s(t)$$

- Simulink representation of the electrical circuit



Continuous & Discrete-Time Models

Continuous-Time Systems

- Most continuous time systems represent how continuous signals are transformed via **differential equations**. e.g. circuit, car velocity

$$\frac{dv_c(t)}{dt} + \frac{1}{RC} v_c(t) = \frac{1}{RC} v_s(t)$$

$$m \frac{dv(t)}{dt} + \rho v(t) = f(t)$$

First order differential equations

Discrete-Time Systems

- Most discrete time systems represent how discrete signals are transformed via **difference equations** e.g. bank account, discrete car velocity system

$$y[n] = 1.01y[n-1] + x[n]$$

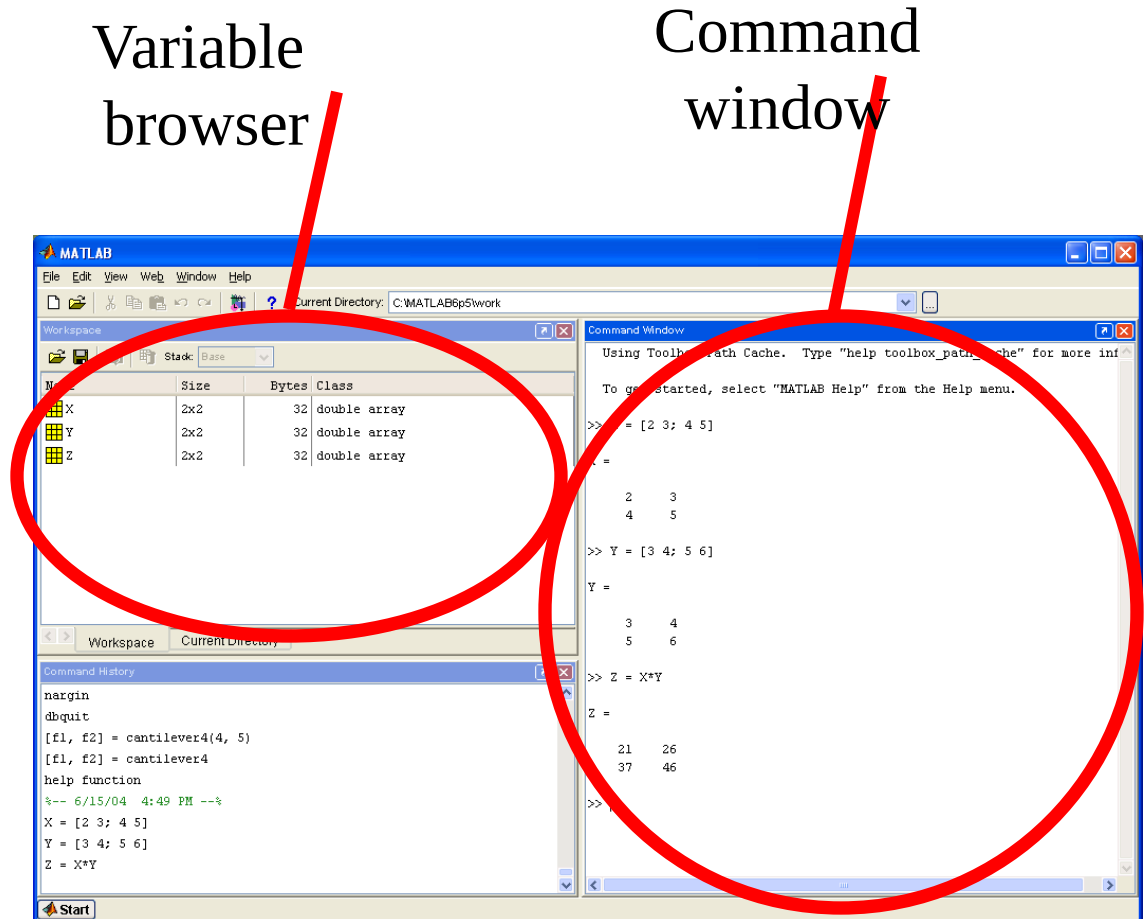
First order difference equations

Properties of a System

- **Causal:** a system is causal if the output at a time, only depends on input values up to that time.
- **Linear:** a system is linear if the output of the scaled sum of two input signals is the equivalent scaled sum of outputs
- **Time-invariance:** a system is time invariant if the system's output is the same, given the same input signal, regardless of time.

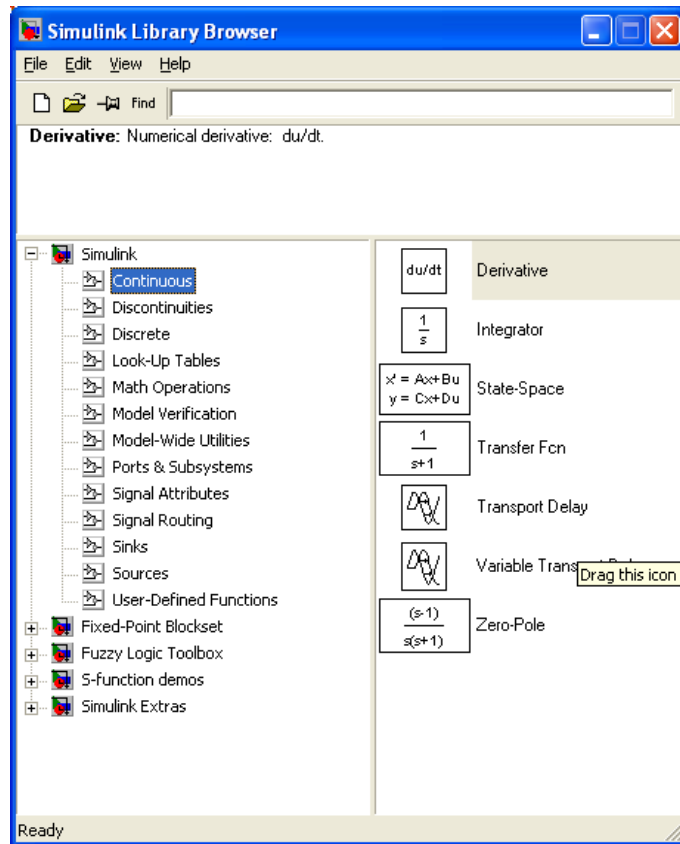
Introduction to Matlab/Simulink (1)

- Click on the Matlab icon/start menu initialises the Matlab environment:
- The main window is the dynamic command interpreter which allows the user to issue Matlab commands
- The variable browser shows which variables currently exist in the workspace



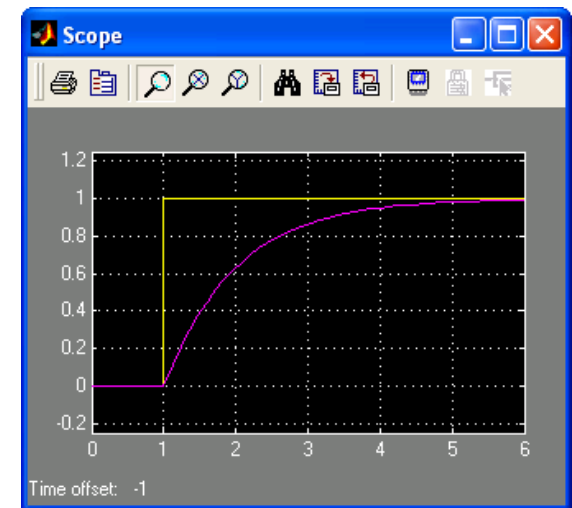
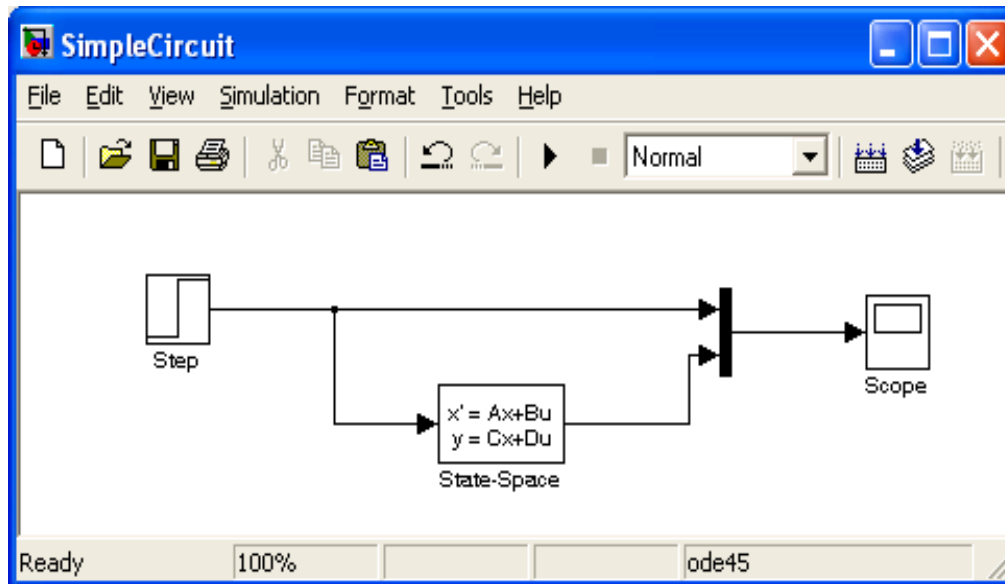
Introduction to Matlab/Simulink (2)

- Type the following at the Matlab command prompt
`>> simulink`
- The following **Simulink library** should appear



Introduction to Matlab/Simulink (3)

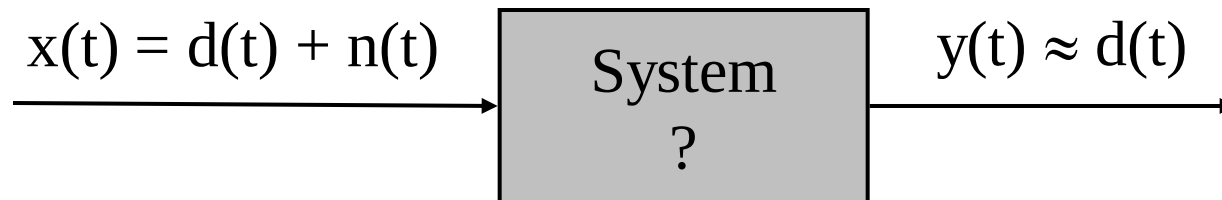
- Click File-New to create a new **workspace**, and drag and drop objects from the library onto the workspace.



- Selecting **Simulation-Start** from the pull down menu will run the dynamic simulation. Click on the blocks to view the data or alter the run-time parameters

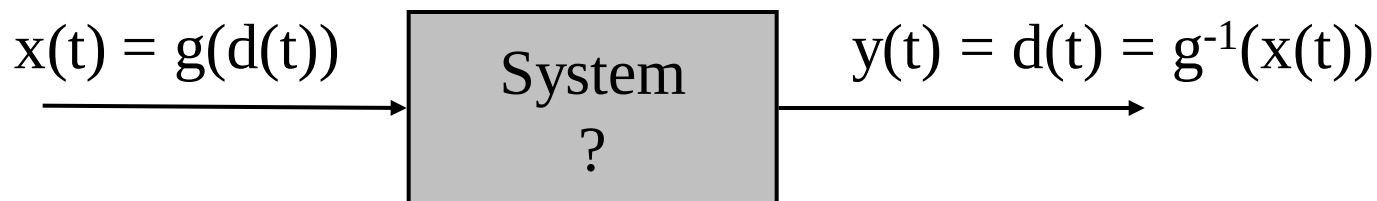
How Are Signal & Systems Related?

- How to design a system to process a signal in particular ways?
- Design a system to restore or enhance a particular signal
 - Remove **high frequency** background communication noise
 - Enhance **noisy** images from spacecraft
- Assume a signal is represented as
 - $x(t) = d(t) + n(t)$
- Design a system to remove the unknown “noise” component $n(t)$, so that $y(t) \approx d(t)$



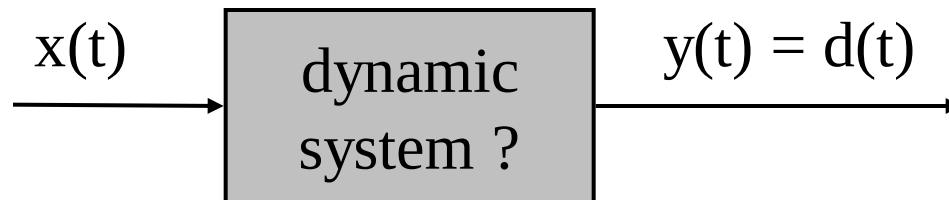
How Are Signal & Systems Related?

- How to design a system to extract specific pieces of information from signals
 - Estimate the heart rate from an electrocardiogram
 - Estimate economic indicators (bear, bull) from stock market values
- Assume a signal is represented as: $x(t) = g(d(t))$
- Design a system to “invert” the transformation $g()$, so that $y(t) = d(t)$



How Are Signal & Systems Related?

- How to design a (dynamic) system to modify or control the output of another (dynamic) system
 - Control an aircraft's altitude, velocity, heading by adjusting throttle, rudder, ailerons
 - Control the temperature of a building by adjusting the heating/cooling energy flow.
- Assume a signal is represented as: $x(t) = g(d(t))$
- Design a system to “invert” the transformation $g()$, so that $y(t) = d(t)$



Lecture 1: Summary

- Signals and systems are important for:
 - Electrical circuits
 - Physical models and control systems
 - Digital media (music, voice, photos, video)
- Study of signals and systems helps in:
 - Design systems to remove noise/enhance measurement from audio and picture/video data
 - Investigate stability of physical structures
 - Control the performance mechanical and electrical devices
- This will be the foundation for studying systems and signals as a generic subject on this course.

Lecture 1: Exercises

- Read Text Chapter 1. (This contains the material for lectures 1-3)
- Questions 1.1, 1.2, 1.4, 1.5, 1.6
- In lecture 2, we'll look at signals in more depth and look at how they can be represented in Matlab/Simulink